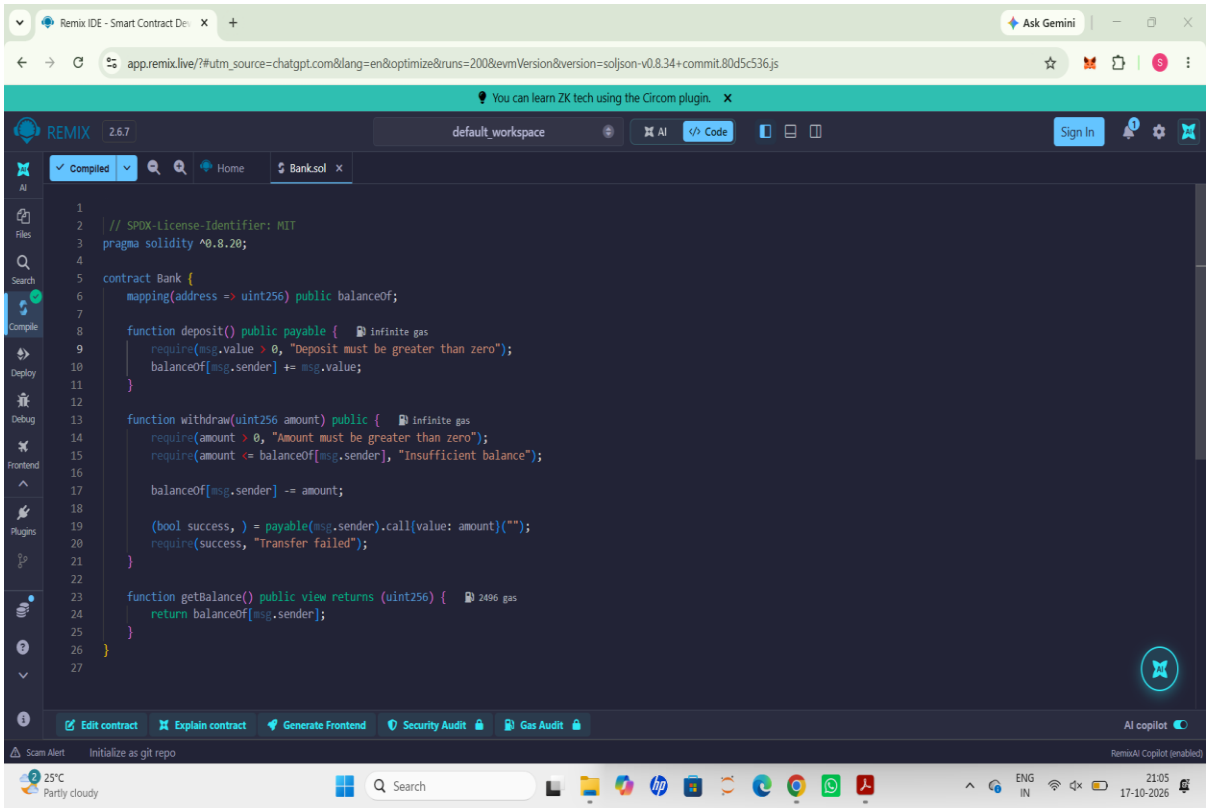
Practical No. 15

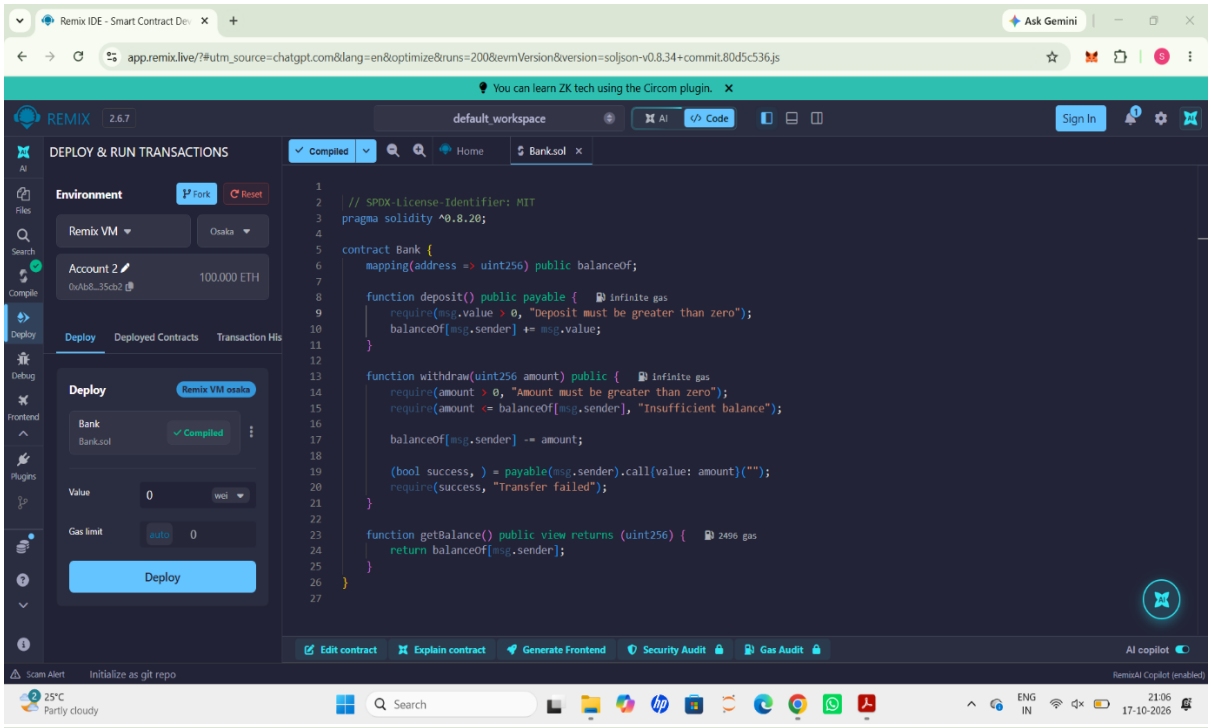
Title: Write a smart contract on a test network, for Bank account of a customer for following operations: Deposit money Withdraw Money Show balance

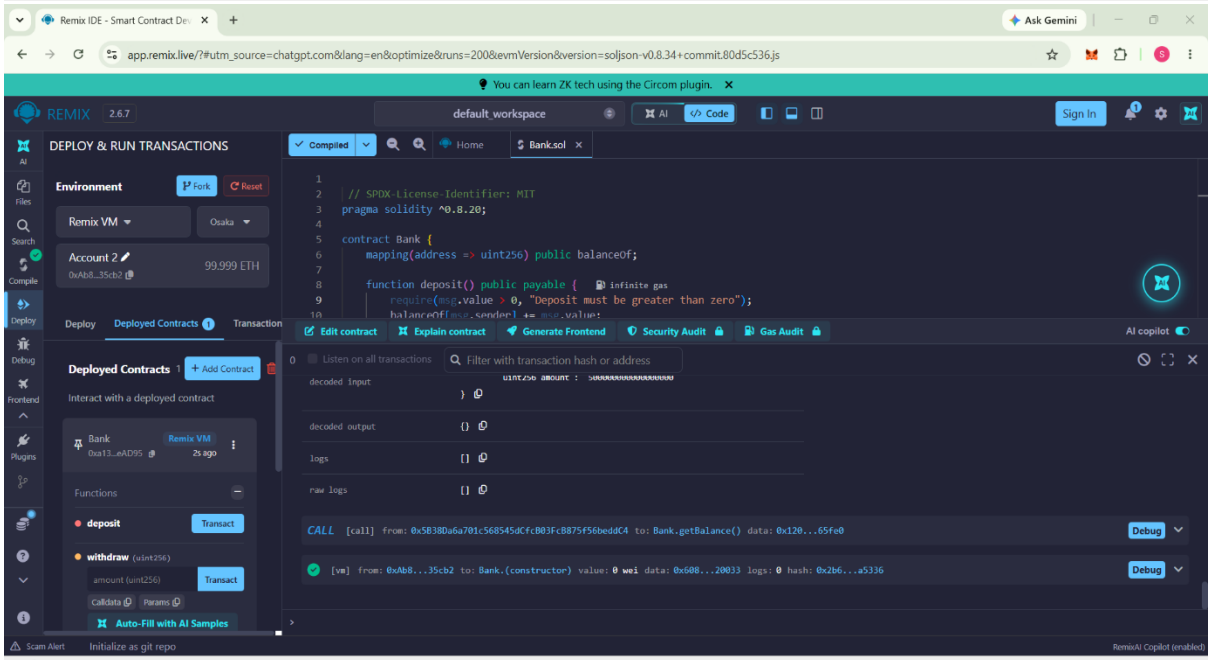
Input:

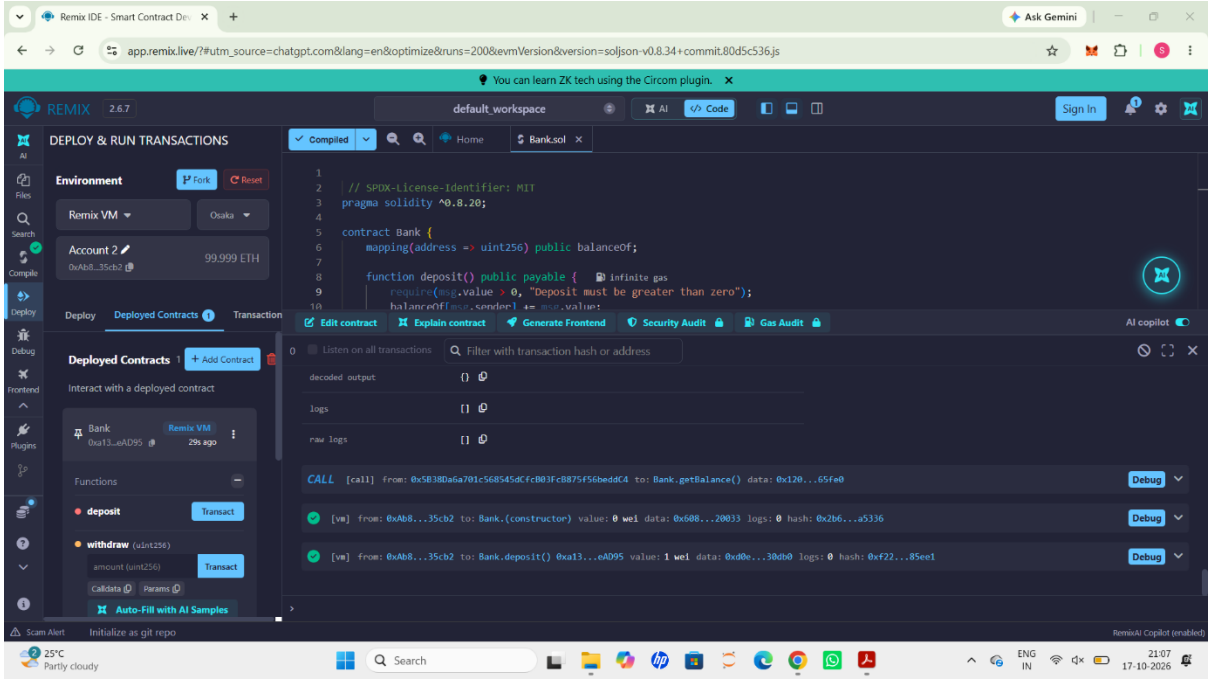
```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract Bank {
    mapping(address => uint256) public balanceOf;
    function deposit() public payable {
        require(msg.value > 0, "Deposit must be greater than zero");
        balanceOf[msg.sender] += msg.value;
    }
    function withdraw(uint256 amount) public {
        require(amount > 0, "Amount must be greater than zero");
        require(amount <= balanceOf[msg.sender], "Insufficient balance");
        balanceOf[msg.sender] -= amount;
        (bool success, ) = payable(msg.sender).call{value: amount}("");
        require(success, "Transfer failed");
    }
    function getBalance() public view returns (uint256) {
        return balanceOf[msg.sender];
    }
}
```

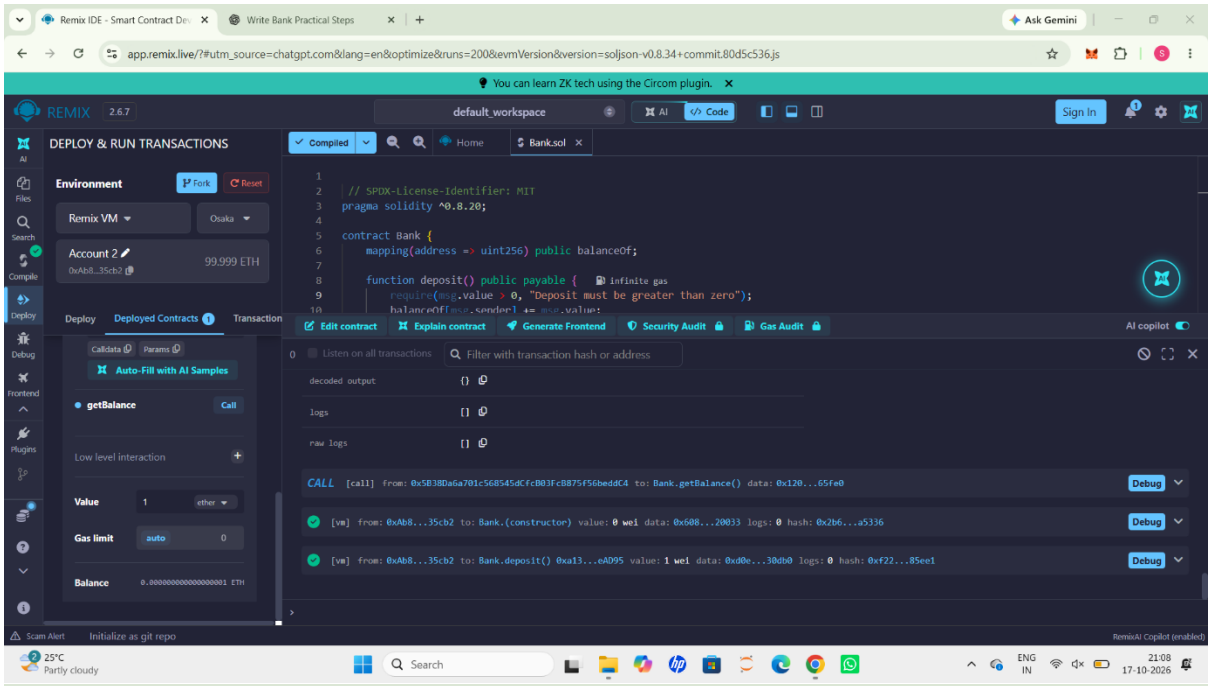
Output:

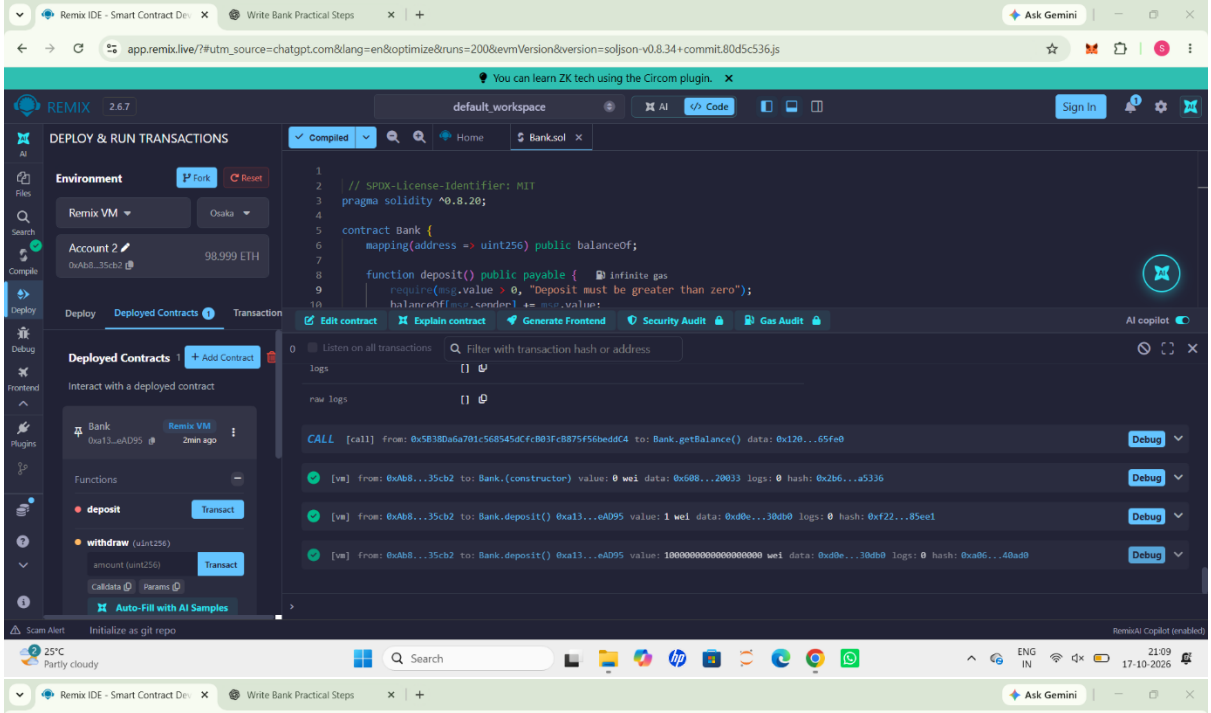


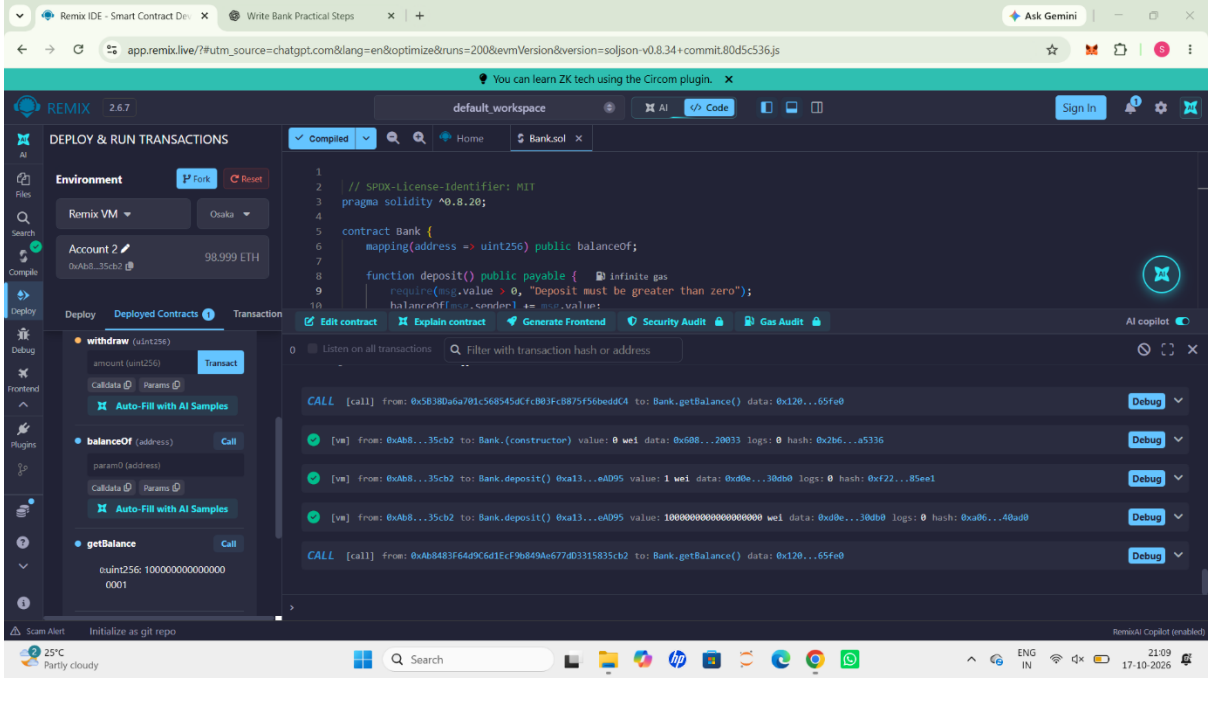

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REMIX 2.6.7
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DEPLOY & RUN TRANSACTIONS
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Osaka
Account 2
0xAb8...35cb2
100.000 ETH
Deploy
Deployed Contracts
Transaction His
Deploy
Remix VM osaka
Bank
Bank.sol
Compiled
Value
0
wei
Gas limit
auto
0
Deploy
Compiled
Home
Bank.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract Bank {
mapping(address => uint256) public balanceOf;
function deposit() public payable { infinite gas
require(msg.value > 0, "Deposit must be greater than zero");
balanceOf[msg.sender] += msg.value;
}
function withdraw(uint256 amount) public { infinite gas
require(amount > 0, "Amount must be greater than zero");
require(amount <= balanceOf[msg.sender], "Insufficient balance");
balanceOf[msg.sender] -= amount;
(bool success,) = payable(msg.sender).call{value: amount}("");
require(success, "Transfer failed");
}
function getBalance() public view returns (uint256) { 2496 gas
return balanceOf[msg.sender];
}
}
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)
25°C
Partly cloudy
Search
ENG
IN
21:06
17-10-2026

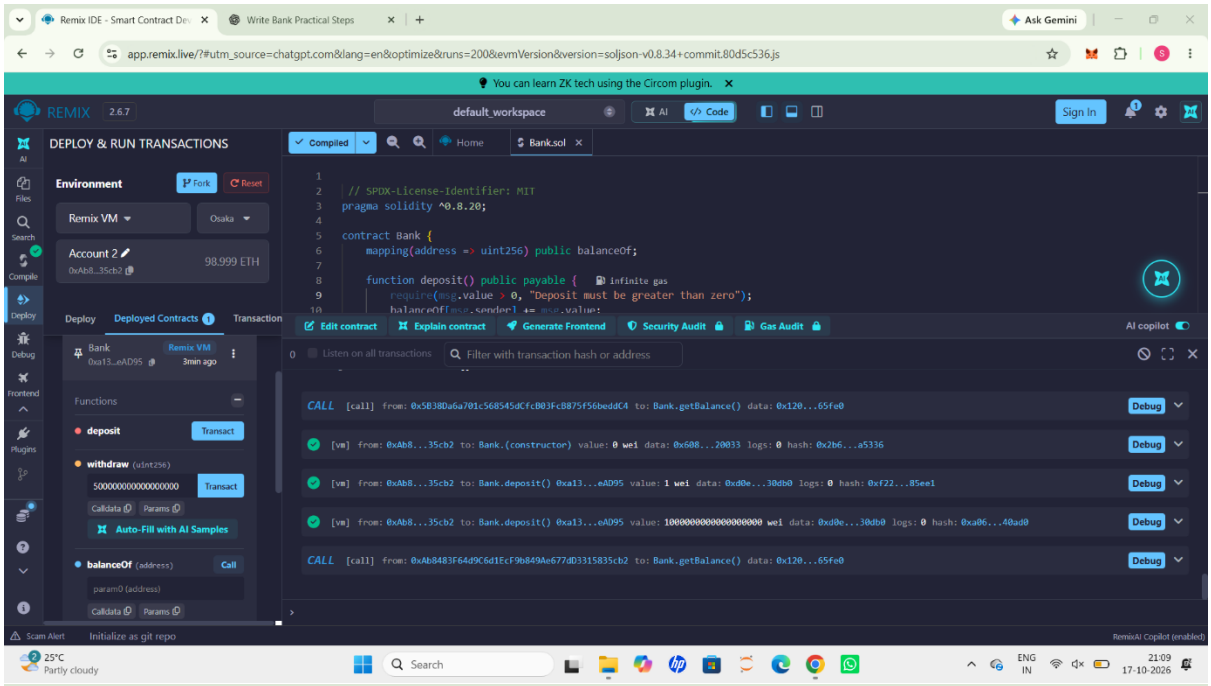

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0xAb8...35cb2
99.999 ETH
Deploy
Deployed Contracts
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Deployed Contracts
Add Contract
Interact with a deployed contract
Bank
Remix VM
0xa13...eAD95
2s ago
Functions
deposit
Transact
withdraw (uint256)
amount (uint256)
Transact
Calldata
Params
Auto-Fill with AI Samples
Compiled
Home
Bank.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract Bank {
mapping(address => uint256) public balanceOf;
function deposit() public payable { infinite gas
require(msg.value > 0, "Deposit must be greater than zero");
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
decoded input
decoded output
{}
logs
[]
raw logs
[]
CALL [call] from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 to: Bank.getBalance() data: 0x120...65fe0
Debug
[vm] from: 0xAb8...35cb2 to: Bank.(constructor) value: 0 wei data: 0x608...20033 logs: 0 hash: 0x2b6...a5336
Debug
Scam Alert
Initialize as git repo
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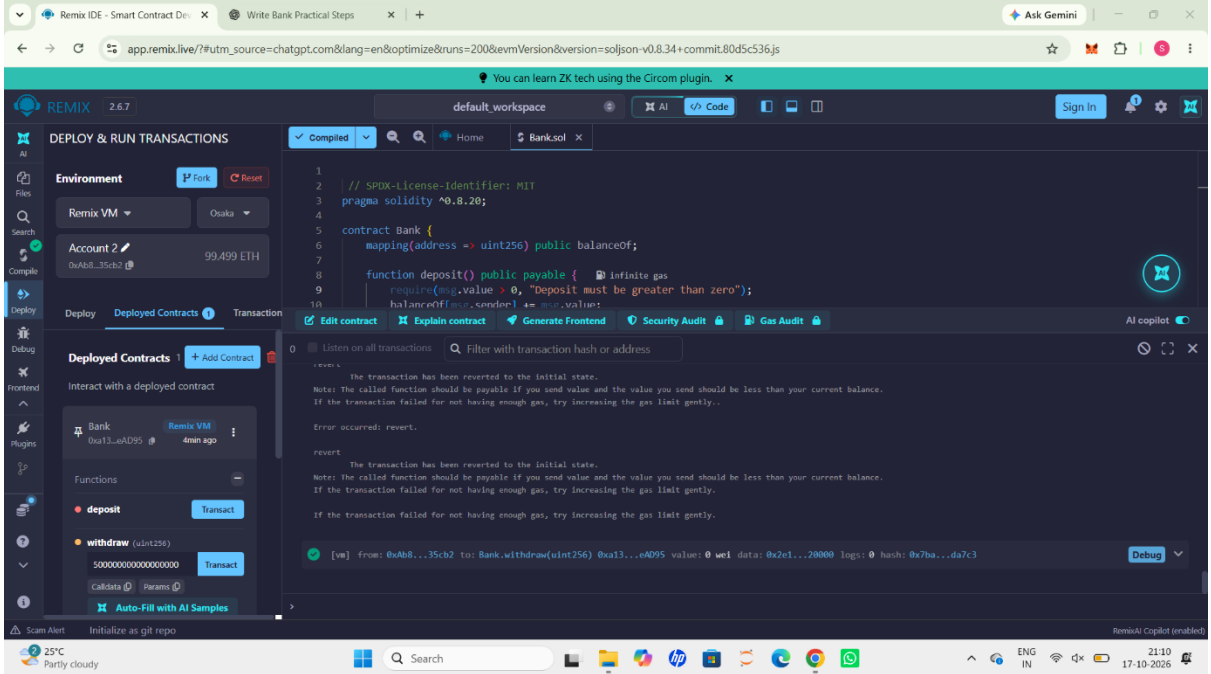

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0xAb8...35cb2
99.999 ETH
Deploy
Deployed Contracts
Transaction
Deployed Contracts
Add Contract
Interact with a deployed contract
Bank
Remix VM
0xa13...eAD95
29s ago
Functions
deposit
Transact
withdraw (uint256)
amount (uint256)
Transact
Calldata
Params
Auto-Fill with AI Samples
Compiled
Home
Bank.sol
// SPDX-License-Identifier: MIT
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Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
decoded output
{}
logs
[]
raw logs
[]
CALL [call] from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 to: Bank.getBalance() data: 0x120...65fe0
Debug
[vm] from: 0xAb8...35cb2 to: Bank.(constructor) value: 0 wei data: 0x608...20033 logs: 0 hash: 0x2b6...a5336
Debug
[vm] from: 0xAb8...35cb2 to: Bank.deposit() 0xa13...eAD95 value: 1 wei data: 0xd0e...30db0 logs: 0 hash: 0xf22...85ee1
Debug
Scam Alert
Initialize as git repo
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Partly cloudy
Search
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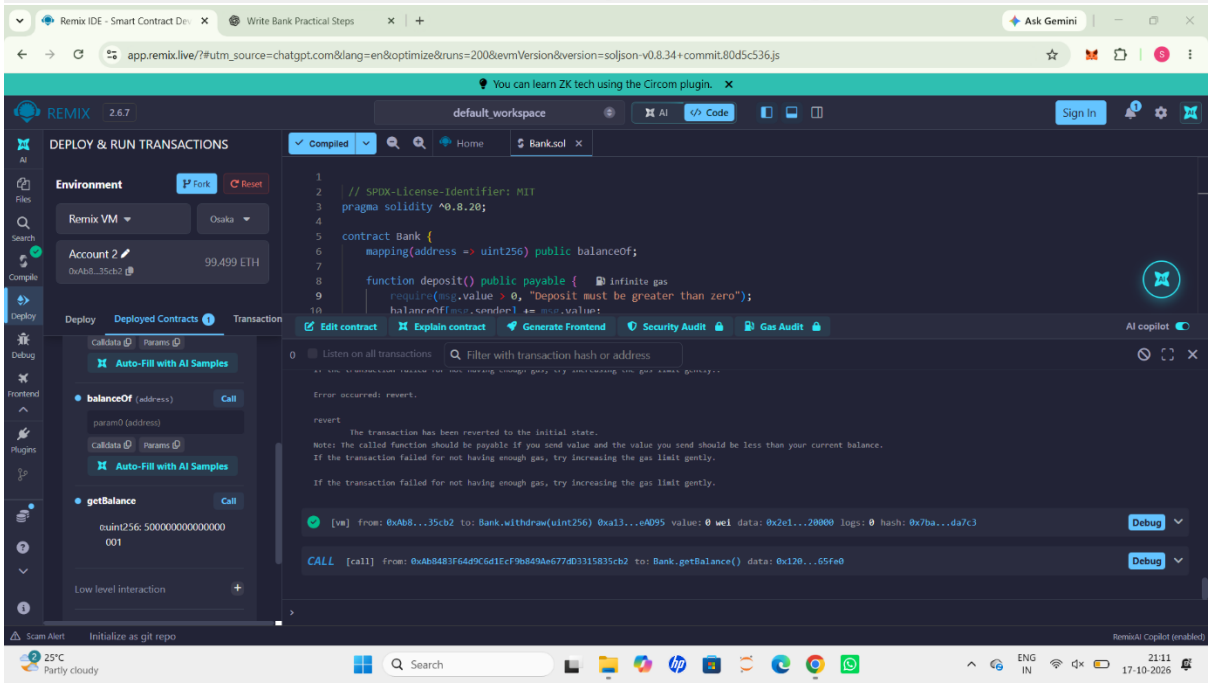

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Osaka
Account 2
0xAb8...35cb2
99.999 ETH
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Deployed Contracts
Transaction
Auto-Fill with AI Samples
getBalance
Call
Low level interaction
Value
Gas limit
Balance
Compiled
Home
Bank.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract Bank {
mapping(address => uint256) public balanceOf;
function deposit() public payable {
require(msg.value > 0, "Deposit must be greater than zero");
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Filter with transaction hash or address
decoded output
logs
raw logs
CALL [call] from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 to: Bank.getBalance() data: 0x120...65fe0
[vm] from: 0xAb8...35cb2 to: Bank.(constructor) value: 0 wei data: 0x608...20033 logs: 0 hash: 0x2b6...a5336
[vm] from: 0xAb8...35cb2 to: Bank.deposit() 0xa13...eAD95 value: 1 wei data: 0xd0e...30db0 logs: 0 hash: 0xf22...85ee1
Debug
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)
25°C
Partly cloudy
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Environment
Remix VM
Osaka
Account 2
0xAb8...35cb2
98.999 ETH
Deploy
Deployed Contracts
Transaction
Deployed Contracts 1
Add Contract
Interact with a deployed contract
Bank
Remix VM
0xa13...eAD95
2min ago
Functions
deposit
Transact
withdraw (uint256)
amount (uint256)
Auto-Fill with AI Samples
Compiled
Home
Bank.sol
// SPDX-License-Identifier: MIT
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contract Bank {
mapping(address => uint256) public balanceOf;
function deposit() public payable {
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Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Filter with transaction hash or address
logs
raw logs
CALL [call] from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 to: Bank.getBalance() data: 0x120...65fe0
[vm] from: 0xAb8...35cb2 to: Bank.(constructor) value: 0 wei data: 0x608...20033 logs: 0 hash: 0x2b6...a5336
[vm] from: 0xAb8...35cb2 to: Bank.deposit() 0xa13...eAD95 value: 1 wei data: 0xd0e...30db0 logs: 0 hash: 0xf22...85ee1
[vm] from: 0xAb8...35cb2 to: Bank.deposit() 0xa13...eAD95 value: 10000000000000000000 wei data: 0xd0e...30db0 logs: 0 hash: 0xa06...40ad0
Debug
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)
25°C
Partly cloudy
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IN
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DEPLOY & RUN TRANSACTIONS
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Osaka
Account 2
0xAb8...35cb2
98.999 ETH
Deploy
Deployed Contracts
Transaction
withdraw (uint256)
amount (uint256)
Transact
Auto-Fill with AI Samples
balanceOf (address)
Call
getBalance
0uint256: 100000000000000001
Compiled
Home
Bank.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract Bank {
mapping(address => uint256) public balanceOf;
function deposit() public payable {
require(msg.value > 0, "Deposit must be greater than zero");
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Filter with transaction hash or address
CALL [call] from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 to: Bank.getBalance() data: 0x120...65fe0
[vm] from: 0xAb8...35cb2 to: Bank.(constructor) value: 0 wei data: 0x608...20033 logs: 0 hash: 0x2b6...a5336
[vm] from: 0xAb8...35cb2 to: Bank.deposit() 0xa13...eAD95 value: 1 wei data: 0xd0e...30db0 logs: 0 hash: 0xf22...85ee1
[vm] from: 0xAb8...35cb2 to: Bank.deposit() 0xa13...eAD95 value: 10000000000000000000 wei data: 0xd0e...30db0 logs: 0 hash: 0xa06...40ad0
CALL [call] from: 0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2 to: Bank.getBalance() data: 0x120...65fe0
Debug
Scam Alert
Initialize as git repo
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Osaka
Account 2
0xAb8...35cb2
98.999 ETH
Deploy
Deployed Contracts
Transaction
Bank
Remix VM
0xa13...eAD95
3min ago
Functions
deposit
Transact
withdraw (uint256)
50000000000000000
Transact
Calldata
Params
Auto-Fill with AI Samples
balanceOf (address)
Call
param0 (address)
Compiled
Home
Bank.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract Bank {
mapping(address => uint256) public balanceOf;
function deposit() public payable {
infinite gas
require(msg.value > 0, "Deposit must be greater than zero");
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
CALL [call] from: 0x5B38Da6a701c568545dCfcB03FcB875f56beddC4 to: Bank.getBalance() data: 0x120...65fe0
Debug
[vm] from: 0xAb8...35cb2 to: Bank.(constructor) value: 0 wei data: 0x608...20033 logs: 0 hash: 0x2b6...a5336
[vm] from: 0xAb8...35cb2 to: Bank.deposit() 0xa13...eAD95 value: 1 wei data: 0xd0e...30db0 logs: 0 hash: 0xf22...85ee1
[vm] from: 0xAb8...35cb2 to: Bank.deposit() 0xa13...eAD95 value: 10000000000000000 wei data: 0xd0e...30db0 logs: 0 hash: 0xa06...40ad0
CALL [call] from: 0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2 to: Bank.getBalance() data: 0x120...65fe0
Scam Alert
Initialize as git repo
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21:09
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Remix VM
Osaka
Account 2
0xAb8...35cb2
99.499 ETH
Deploy
Deployed Contracts
Transaction
Deployed Contracts 1
+ Add Contract
Interact with a deployed contract
Bank
Remix VM
0xa13...eAD95
4min ago
Functions
deposit
Transact
withdraw (uint256)
50000000000000000
Transact
Calldata
Params
Auto-Fill with AI Samples
Compiled
Home
Bank.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract Bank {
mapping(address => uint256) public balanceOf;
function deposit() public payable {
infinite gas
require(msg.value > 0, "Deposit must be greater than zero");
Edit contract
Explain contract
Generate Frontend
Security Audit
Gas Audit
AI copilot
Listen on all transactions
Filter with transaction hash or address
The transaction has been reverted to the initial state.
Note: The called function should be payable if you send value and the value you send should be less than your current balance.
If the transaction failed for not having enough gas, try increasing the gas limit gently..
Error occurred: revert.
revert
The transaction has been reverted to the initial state.
Note: The called function should be payable if you send value and the value you send should be less than your current balance.
If the transaction failed for not having enough gas, try increasing the gas limit gently.
If the transaction failed for not having enough gas, try increasing the gas limit gently.
[vm] from: 0xAb8...35cb2 to: Bank.withdraw(uint256) 0xa13...eAD95 value: 0 wei data: 0x2e1...20000 logs: 0 hash: 0x7ba...da7c3
Debug
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)
25°C
Partly cloudy
Search
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IN
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Osaka
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0xAb8...35cb2
99.499 ETH
Deploy
Deployed Contracts
Transaction
Auto-Fill with AI Samples
balanceOf (address)
Call
param0 (address)
getBalance
Call
uint256: 500000000000000001
Low level interaction
Compiled
Home
Bank.sol
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract Bank {
mapping(address => uint256) public balanceOf;
function deposit() public payable {
infinite gas
require(msg.value > 0, "Deposit must be greater than zero");
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[vm] from: 0xAb8...35cb2 to: Bank.withdraw(uint256) 0xa13...eAD95 value: 0 wei data: 0x2e1...20000 logs: 0 hash: 0x7ba...da7c3
Debug
CALL [call] from: 0xAb8483F64d9C6d1EcF9b849Ae677dD3315835cb2 to: Bank.getBalance() data: 0x120...65fe0
Debug
Scam Alert
Initialize as git repo
RemixAI Copilot (enabled)
25°C
Partly cloudy
Search
ENG
IN
21:11
17-10-2026